RIDEWISE

Ride-Hailing Customer Analytics and Churn Prediction

We predict which riders are about to stop using the app, so the team can win them back before they go.

1. Get the data ready
2. Turn behaviour into numbers
3. Teach a model to spot leavers
4. Turn it into a plan that saves money

THE BIG PICTURE

What are we actually trying to do?

The problem

Nearly 1 in 3 riders quietly stops using RideWise each quarter. By the time it shows up in a report, they are already gone — and winning back a lost customer is far harder than keeping one.

The goal

Spot the riders who are about to leave — while they are still here — so the team can send a well-timed offer to the right people, instead of paying everyone.

“Churn” — the one word worth knowing

Churn just means a customer leaving. A **“churn model”** is a tool that looks at how someone has behaved and estimates how likely they are to leave soon. That estimate is the single thing this whole project produces — everything else is about making it accurate, then acting on it.

Riders → their behaviour → a model → a “who’s likely to leave” list → a targeted offer

STEP 1 · GETTING THE DATA READY

Five spreadsheets, checked and tidied

10,000	200,000	50,000	5,000	20
Riders	**Trips**	**Sessions**	**Drivers**	**Promotions**
who they are	every ride taken	app browsing	who drove	past offers

In plain terms

We check the data before trusting it.

Same idea as proof-reading before you send an email. We made sure every ride belongs to a real rider (it did — 100%), and set rules to throw out impossible records: rides with no fare, or lasting longer than a day.

Here, nothing needed removing — the data was already clean. We keep the checks anyway, because real-world data rarely is.

A surprise worth knowing

The data came with a “will they leave?” column already filled in. We tested it — and found it was **random noise**. It had no real connection to how riders behaved.

So we built our own honest measure instead: **did the rider take any trip in the following month?** No trip = they left. That becomes our source of truth.

STEP 2 · TURNING BEHAVIOUR INTO NUMBERS

A model can’t read stories — so we score habits

For each rider we summarise their history into 22 simple numbers. A model can compare numbers; it can’t read a diary of trips. Here are the most important ones:

Days since last ride
Been a while? A warning sign.

How often they ride
Regulars rarely leave.

Total money spent
Big spenders are more loyal.

Ratings they give
Unhappy riders leave more.

Surge prices paid
Frustrating for the rider.

Loyalty tier
Higher tier, more committed.

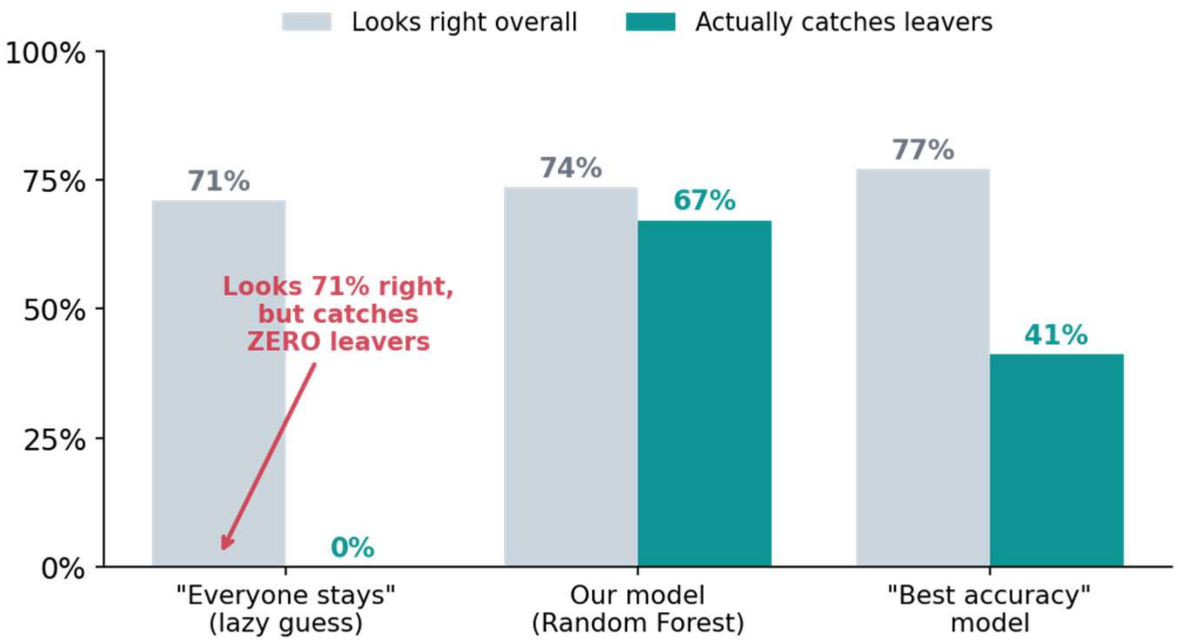
In plain terms

The golden rule: only use the past to predict the future.

We freeze time at a chosen day. We describe each rider using only what happened before that day, then check what they did after. That way the model never accidentally “peeks” at the answer — the mistake that makes a model look brilliant in testing and fail in real life.

STEP 3 · TEACHING THE MODEL

Why “being right a lot” can still be useless



In plain terms

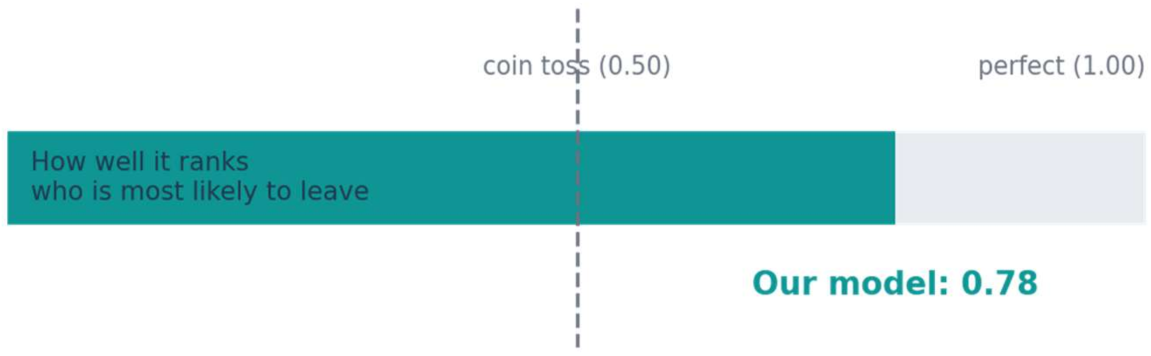
“Accuracy” can lie to you.

Since about 3 in 4 riders stay, a model can just say “everyone stays” and be right 71% of the time — while catching **none** of the people who actually leave. Useless.

So we don’t judge our model on “how often it’s right.” We judge it on **how many real leavers it catches** — the green bars. That’s the number that saves customers.

STEP 3 · HOW GOOD IS IT?

Think of it as ranking riders from safest to riskiest



0.78 out of 1.00

In plain terms

It doesn’t give a yes/no — it ranks.

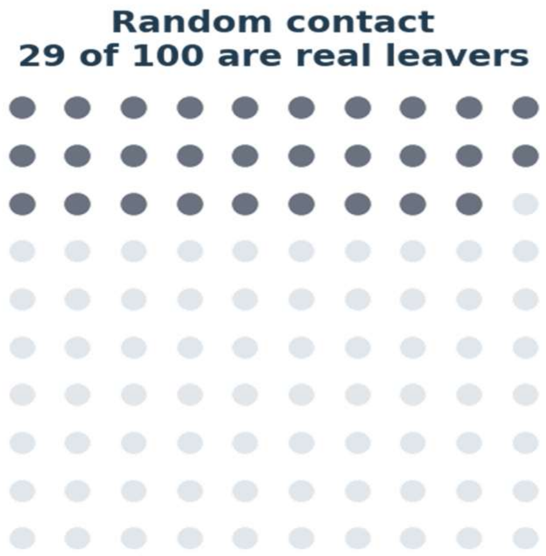
The model lines up all riders from “very safe” to “about to leave.”

We score that ranking from 0.50 (no better than flipping a coin) to 1.00 (perfect). **Ours scores 0.78** — a strong, reliable signal, and two different model types agreed on it, which tells us it’s real and not a fluke.

Two model types agreeing is like two doctors reaching the same diagnosis independently — more trustworthy than one.

STEP 3 · WHY THE LIST IS WORTH USING

If you contact 100 riders, how many are real leavers?



In plain terms

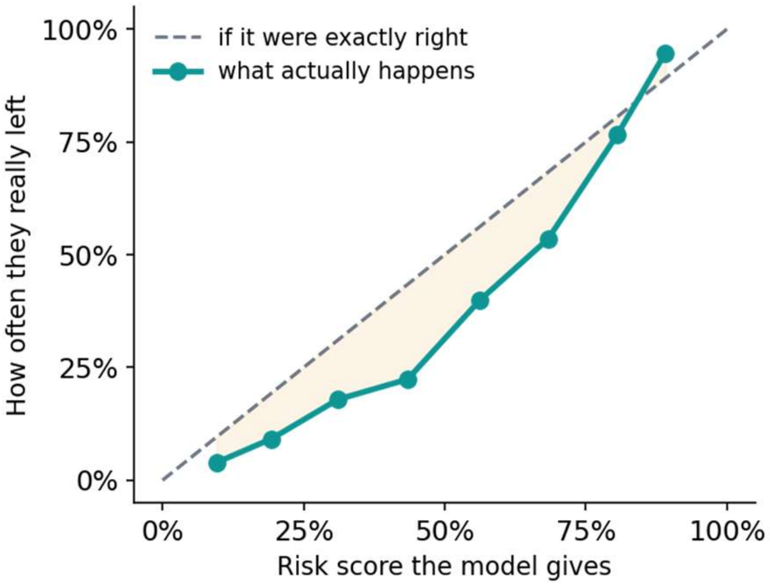
Nearly 8 in 10 are real.

Pick 100 riders at random and about **29** are truly leaving.

Use the model's list and about **79** are — almost **3× better**. Far less budget wasted on people who were staying anyway.

STEP 3 · ONE HONEST LIMITATION

The model worries a bit too much — and that's OK here



In plain terms

When it says 45%, the real rate is nearer 23%.

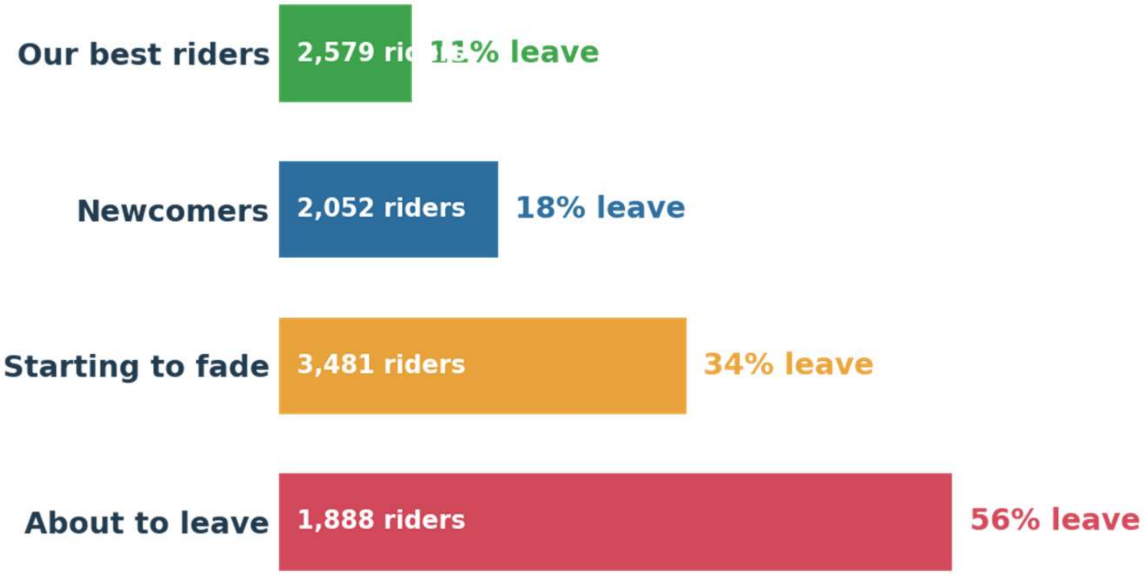
The risk scores run a little high — like a smoke alarm that's quick to beep. The exact percentages aren't perfectly tuned.

In plain terms

Why it doesn't matter for us.

We never use the exact percentage. We only use the **order** — who's riskier than whom — to pick the top slice to contact. The ranking is what we act on, and the ranking is solid. (If we ever needed exact odds, there's a standard one-line fix.)

Four kinds of rider — and only one needs the budget



Our best riders

Loyal & high-spending. Leave them be.

Newcomers

Still forming the habit. Encourage them.

Starting to fade

Watch closely, nudge the slipping ones.

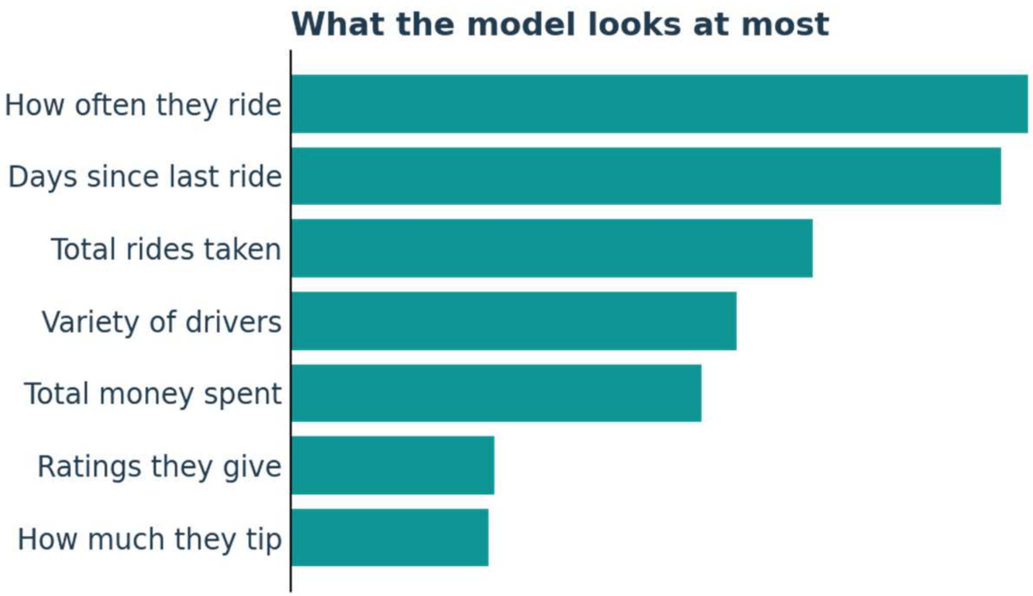
About to leave

Where most churn is. Act here first.

How we chose four groups: the maths alone suggested two, but two is too blunt to act on. Four gives clearly different, useful groups — a judgement call the analyst makes, not the computer.

STEP 3 · CAN WE TRUST IT?

The model learned exactly what common sense expects



It focuses on sensible things.

The biggest clues it uses are how often someone rides and how long since their last trip — exactly what you'd guess. That's reassuring: it learned real patterns, not random quirks.

We can explain any single decision.

For any flagged rider we can show the reasons — e.g. "hasn't ridden in 5 weeks, and hit surge prices often." No mysterious black box.

STEP 4 · FROM A SCORE TO A PLAN

Who do we actually contact?

We can’t contact everyone — so we contact the riskiest 15%.

The model ranks all riders by risk. The team has limited time and budget, so we draw a line and focus on the top 15% most likely to leave. Move the line whenever capacity changes — it’s a business dial, not a fixed rule.

79%	41%	0
of those contacted are truly leaving	of all leavers are caught, contacting just 15%	of our best riders wasted budget on

Then we match the offer to the rider.

High-value riders at risk get a stronger offer (a credit); lower-value ones get a lighter nudge (points). Spend follows the money that’s actually at risk.

STEP 4 · THE PAYOFF

The same offer — but sent to the right people

Offer everyone

$150k

Costs more than it saves. Most of it goes to riders who were never going to leave — a net loss.

Offer the right 15%

$15k

One-tenth the cost, and it protects far more value than it spends. The campaign pays for itself several times over.

90% less

spent on retention offers

$135k

freed up every campaign

pays for itself

several times over

Same offer. Same budget per person. The only change is who receives it — that’s the whole win.

THE WHOLE JOURNEY

Four steps, one sentence each

1	**Get the data ready**	Check five spreadsheets, tidy them, and define “leaving” in a way we can trust.
2	**Turn behaviour into numbers**	Summarise each rider’s habits into 22 numbers — using only the past.
3	**Teach a model to spot leavers**	Judge it on real leavers caught, not raw accuracy. It ranks riders 0.78 out of 1.00.
4	**Turn it into a money-saving plan**	Contact the riskiest 15% with the right offer — 90% cheaper, and it pays for itself.

THE TAKEAWAY

Don’t pay everyone to stay.
Find the ones about to leave, and reach them first.

Catch leavers early

the model ranks who’s at risk

Contact only 15%

the right people, not everyone

Save 90% of the budget

and it pays for itself

That’s RideWise — from raw spreadsheets to a smarter, cheaper way to keep customers.